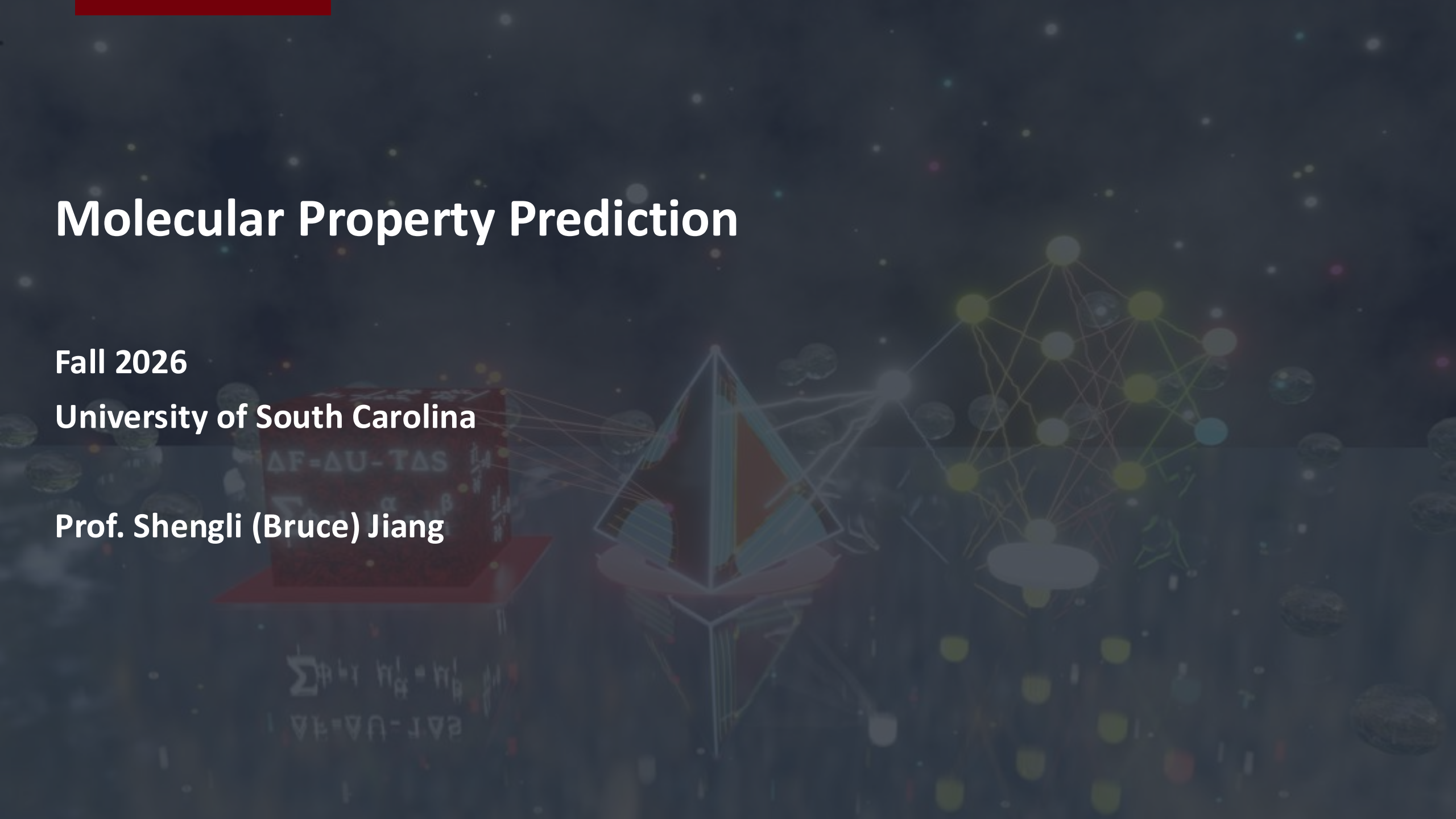
Molecular Property Prediction

Fall 2026

University of South Carolina

Prof. Shengli (Bruce) Jiang



From Lecture 26

Molecular Featurization III: Descriptors

- *Generate and compare Morgan, MACCS, and RDKit fingerprints*
- *Calculate hydrogen-bond counts, TPSA, and calculated logP*
- *Construct a descriptor matrix*

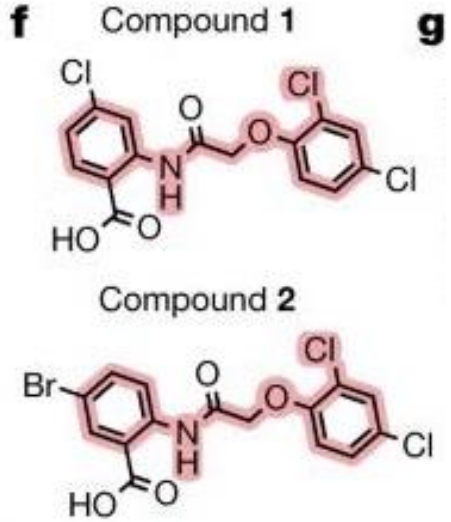
In this lecture, you will learn

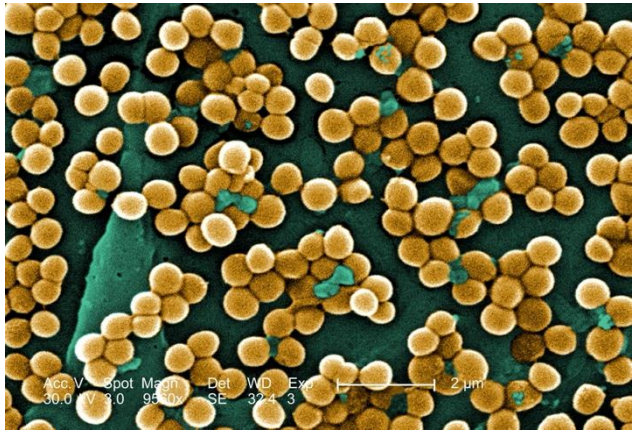
- *Construct molecular inputs and experimental targets*
- *Train a regression model and a classification model*
- *Select model settings using training cross-validation*

Molecular Property Prediction: Discover New Antibiotics



- *Problem:* Many bacteria are becoming resistant to existing antibiotics, making infections harder to treat.
- *Predicted properties:* Whether a molecule can kill bacteria and whether it is toxic to human cells.
- *Approach:* Machine learning (GNN) screened over 12 million molecules to identify promising antibiotics.
- *Success:* Compound 1 reduced MRSA bacterial counts by approximately 94% in mouse skin and thigh infection models, compared with untreated controls.

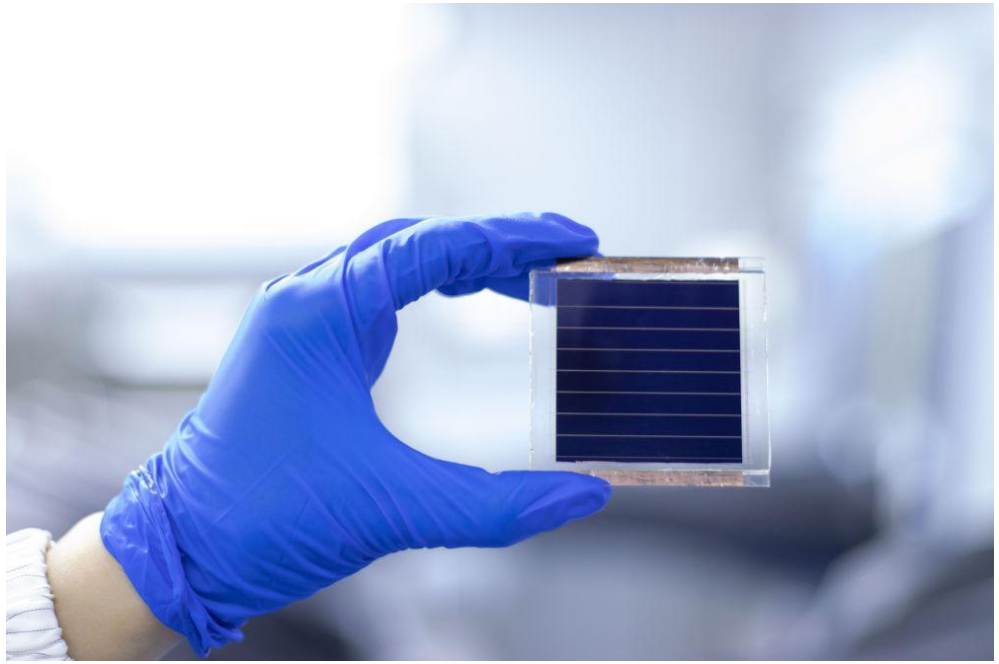




methicillin-resistant Staphylococcus aureus (MRSA)

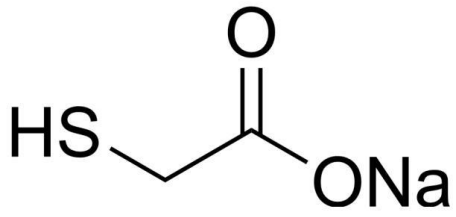
Wong, F.; Zheng, E. J.; Valeri, J. A.; et al. Discovery of a Structural Class of Antibiotics with Explainable Deep Learning. Nature 2024, 626, 177–185. https://doi.org/10.1038/s41586-023-06887-8

Improve Solar Cell Efficiency



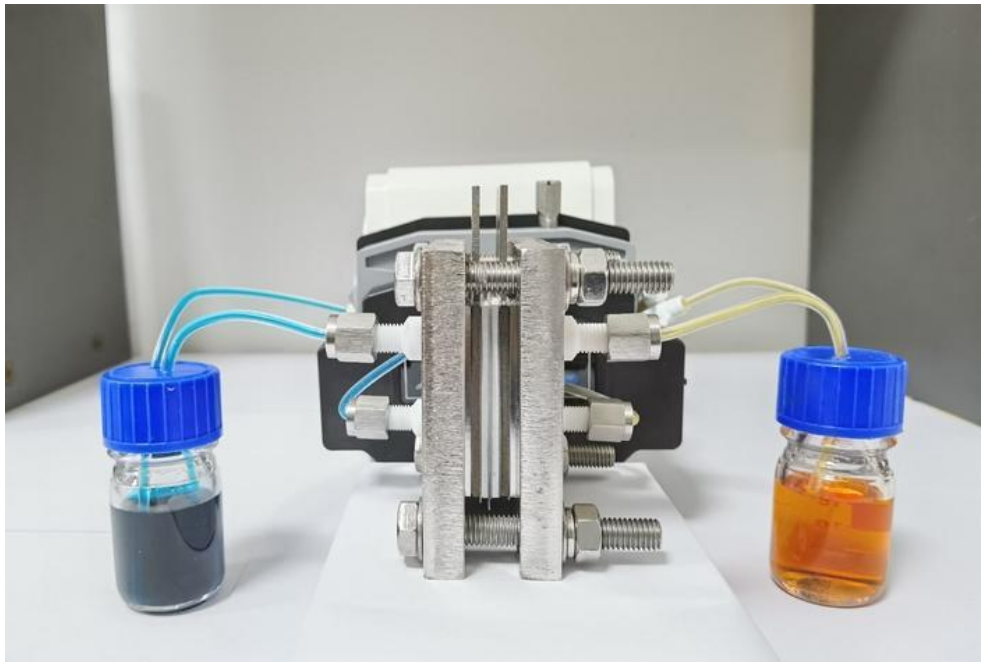
- *Problem:* Tiny defects in solar-cell materials can trap electrical charges and reduce the amount of electricity generated.
- *Predicted properties:* How strongly different molecules attach to the solar-cell surface and help reduce defects.
- *Approach:* Machine learning (Random Forest) helped researchers select molecules for treating the surface.
- *Success:* Without treatment: 22.91% power-conversion efficiency. With ST treatment: 24.56% power-conversion efficiency.

Xu, J.; Chen, H.; Grater, L.; et al. Anion Optimization for Bifunctional Surface Passivation in Perovskite Solar Cells. Nature Materials 2023, 22, 1507–1514. https://doi.org/10.1038/s41563-023-01705-y



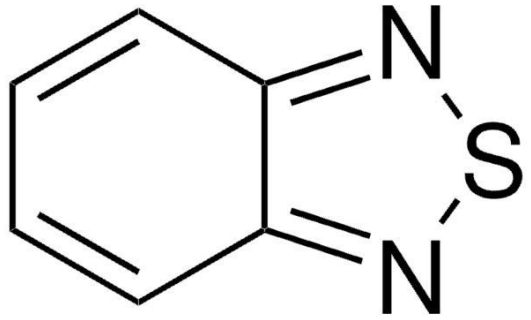
Sodium thioglycolate

Develop Better Battery Electrolyte



- *Problem:* Flow batteries store energy in chemical solutions. Poor solubility limits how much energy-storing material the liquid can hold.
- *Predicted properties:* Solubility, or how much of a molecule can dissolve in different solvent mixtures.
- *Approach:* Machine learning (GPR) predicted promising solvent combinations and guided automated experiments.
- *Success:* Researchers identified 18 promising mixtures after testing only 218 of 2101 possible candidates.

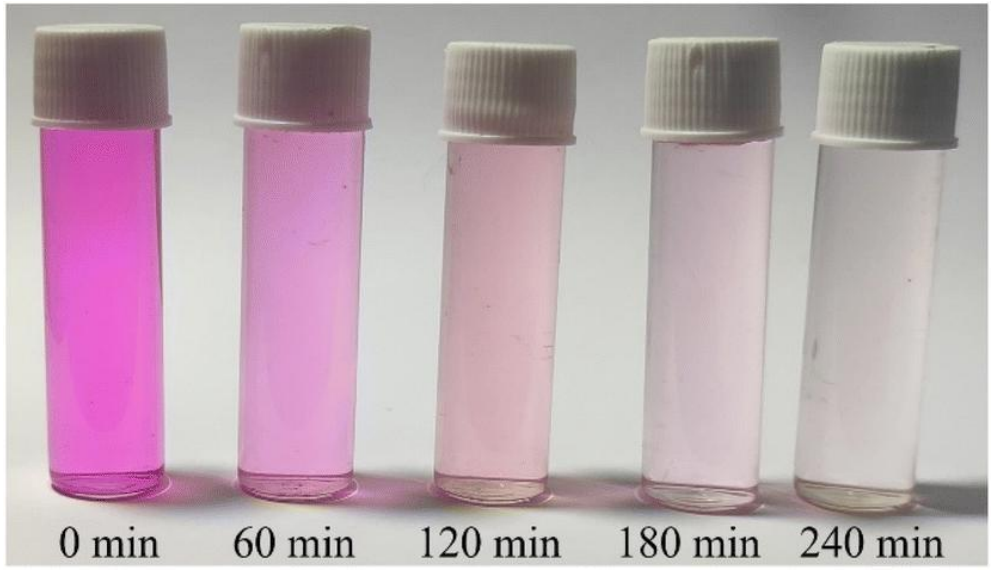
Noh, J.; Doan, H. A.; Job, H.; et al. Nature Communications 2024, 15, 2757. https://doi.org/10.1038/s41467-024-47070-5



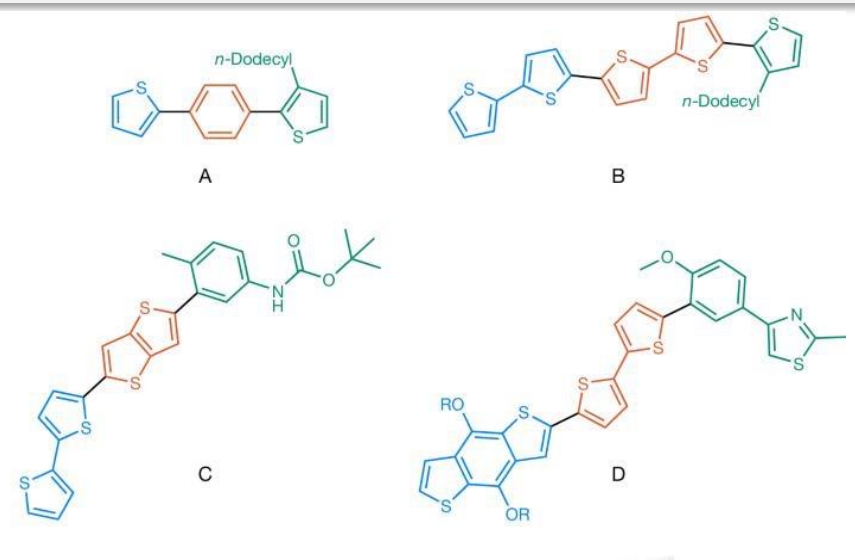
Energy-storing molecule: 2,1,3-benzothiadiazole (BTZ)

Best formulation: 80% 1,4-dioxane + 20% dimethyl sulfoxide (DMSO), by volume. It achieved 6.50 M, approximately 19% higher than pure dioxane.

Design Molecules that Resist Sunlight



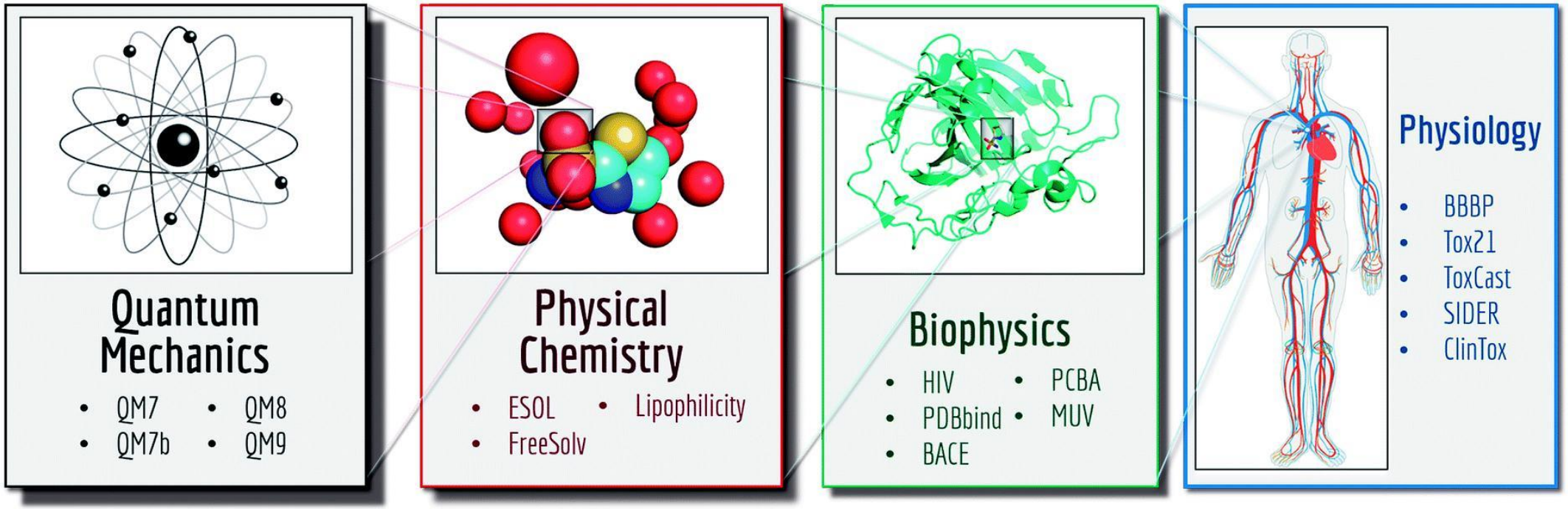
- *Problem:* Some light-absorbing molecules gradually break down when exposed to sunlight, limiting their usefulness.
- *Predicted properties:* Photostability, or how well a molecule resists degradation under light exposure.
- *Approach:* Machine learning (SVM) helped identify molecular structures associated with slower degradation.
- *Success:* Researchers identified molecules with up to four times greater photostability under their experimental conditions.



Angello, N. H.; Friday, D. M.; Hwang, C.; et al. Closed-Loop Transfer Enables Artificial Intelligence to Yield Chemical Knowledge. Nature 2024, 633, 351–358. https://doi.org/10.1038/s41586-024-07892-1.

Regression and Classification Targets

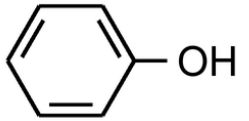
Dataset	What we predict	Task	Molecules
ESOL	How much of a molecule dissolves in water	Regression	1128
BBBP	Whether a molecule can cross the blood-brain barrier	Classification	2053



Wu, Z.; Ramsundar, B.; Feinberg, E. N.; et al. MoleculeNet: A Benchmark for Molecular Machine Learning. Chemical Science 2018, 9, 513–530. https://doi.org/10.1039/C7SC02664A

ESOL Dataset: Predicting Water Solubility

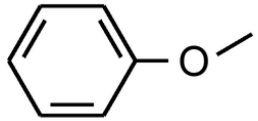
Can we predict how much of a compound dissolves in water from its molecular structure?



Phenol
logS = 0.00



Benzene
logS = −1.64



Anisole
logS = −1.85

- *Dataset*: ESOL contains molecular structures and experimentally measured water solubilities for 1128 compounds.
- *Each row*: one compound, its SMILES string, and its measured **logS**.
- *Model input*, **X:** numerical descriptors or fingerprints calculated from the molecular structure.
- *Prediction target*, **y:** measured logS, a continuous value, so regression task.

$$\log S = \log_{10}\left(\frac{S}{1\ \mathrm{mol\,L^{-1}}}\right)$$

$$\log S = -3 \quad \Rightarrow \quad S = 10^{-3}\ \mathrm{mol\,L^{-1}}$$

Higher logS means greater solubility; one unit higher means ten times greater solubility.

Building the Feature Matrix and Target Vector

What numerical information will the model use to predict logS?

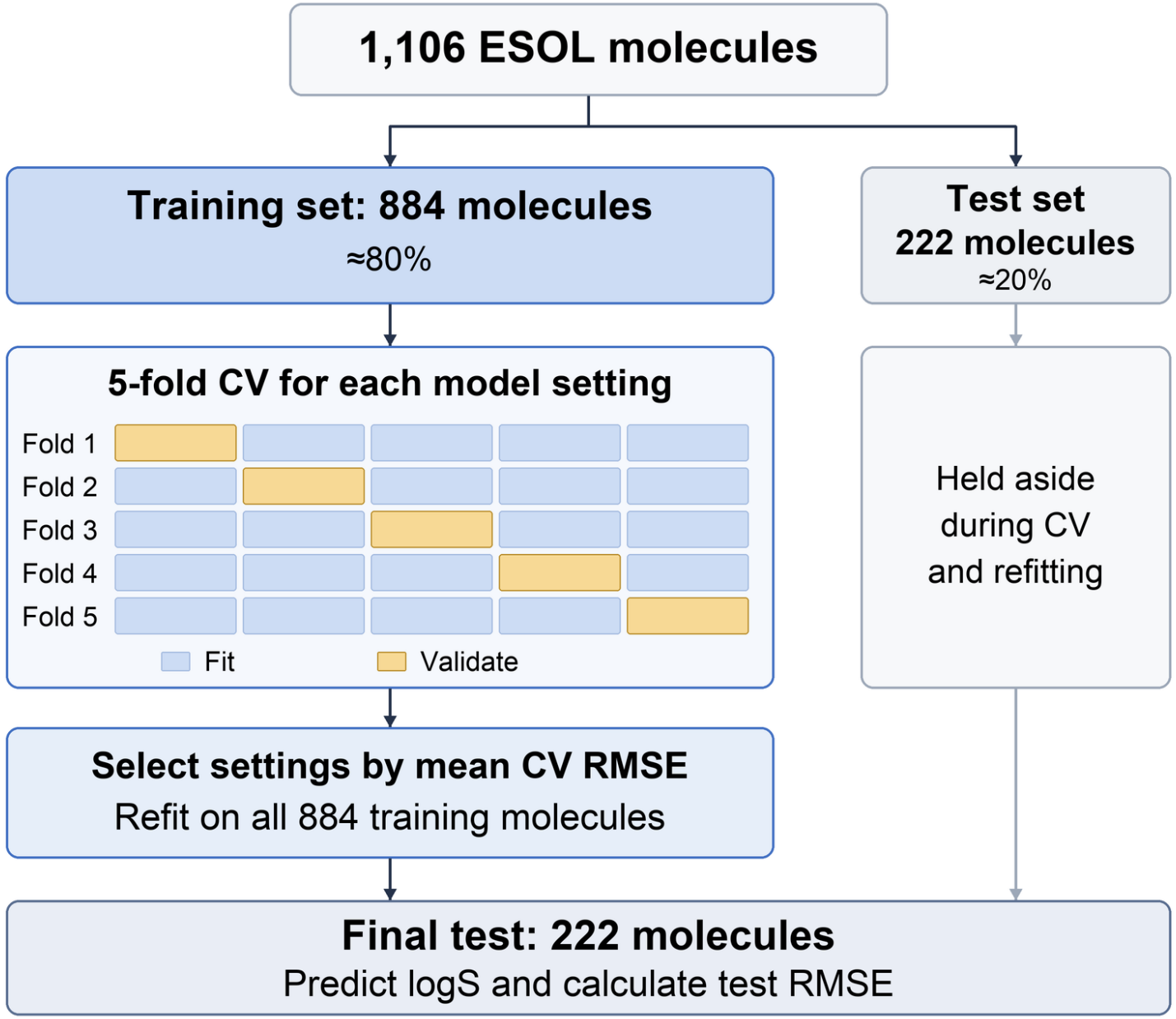
	Feature matrix X					Target y
Molecule	**MW** (g/mol)	**HBD**	**HBA**	**TPSA** (Å²)	**MolLogP**	**Measured logS**
Phenol	94.11	1	1	20.23	1.39	**0.00**
Benzene	78.11	0	0	0.00	1.69	**−1.64**
Anisole	108.14	0	1	9.23	1.70	**−1.85**

Five descriptor columns → X
Measured logS column → y

MW: molecular weight; HBD/HBA: hydrogen-bond donor/acceptor counts; TPSA: topological polar surface area.

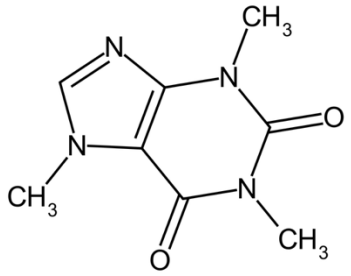
MolLogP is calculated from structure; the logS target comes from experiment.

Five-Fold Cross-Validation



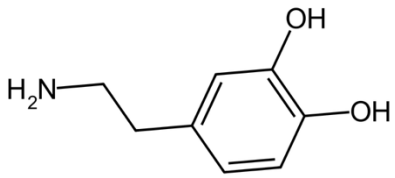
BBBP Dataset: Predicting Blood–Brain Barrier Penetration

Caffeine



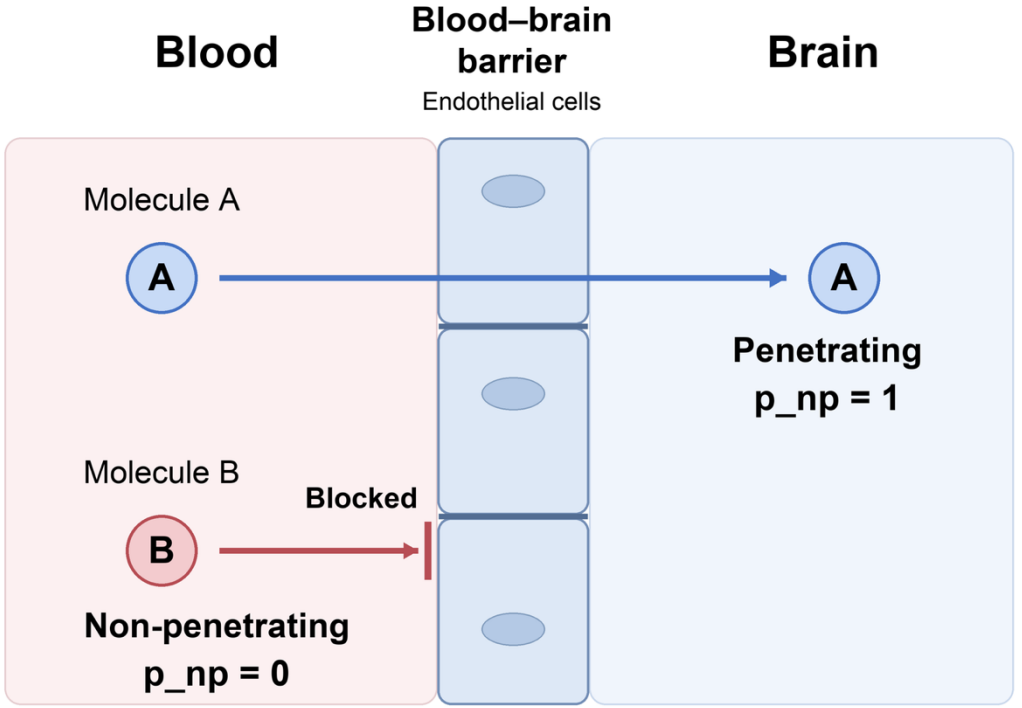
Penetrating
p_np = 1

Dopamine



Non-penetrating
p_np = 0

Can we predict whether a molecule crosses the blood–brain barrier from its molecular structure?



- *Dataset:* BBBP contains molecular structures and binary labels describing blood–brain barrier penetration.
- *Each row:* One compound, its SMILES string, and its penetration label, p_np.
- *Model input, X:* Molecular fingerprints calculated from the structure. We will use 2048-bit Morgan fingerprints.
- *Prediction target, y:* A binary label, making this a classification task.

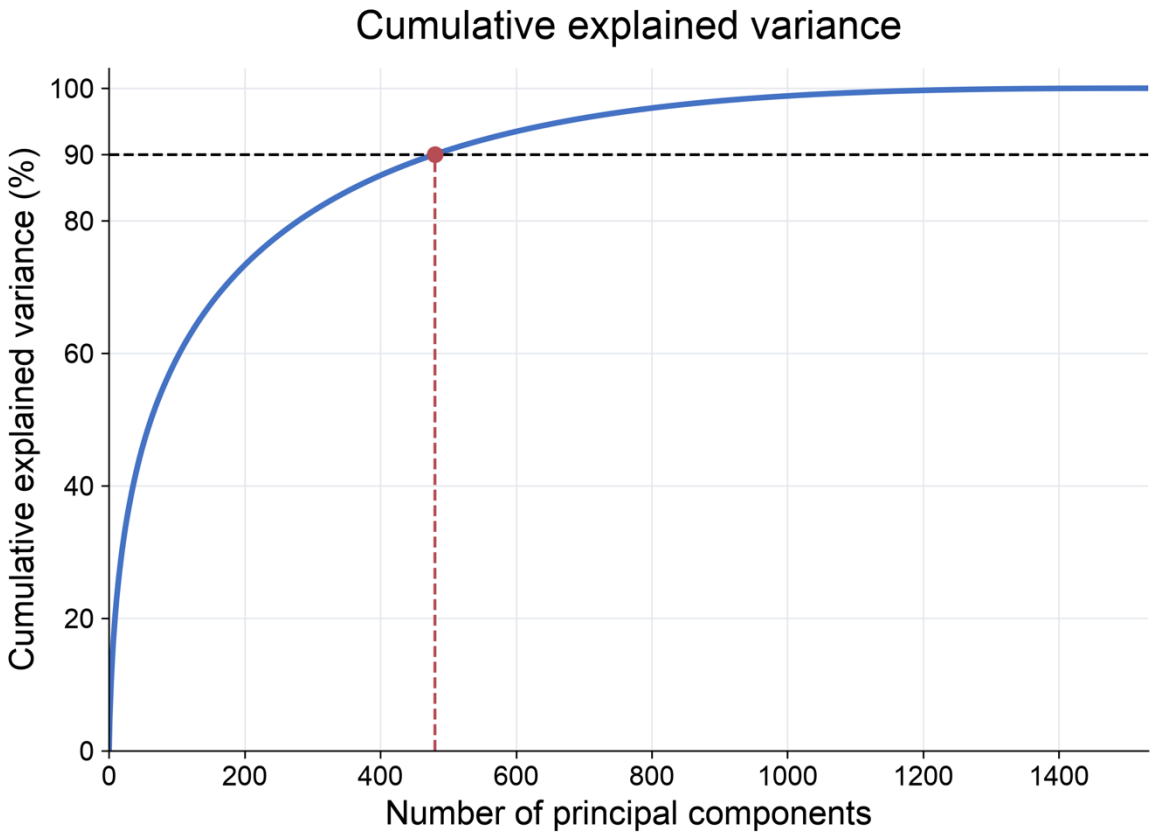
Label	Meaning
1: Penetrating	Classified as crossing the blood–brain barrier
0: Non-penetrating	Classified as not crossing the blood–brain barrier

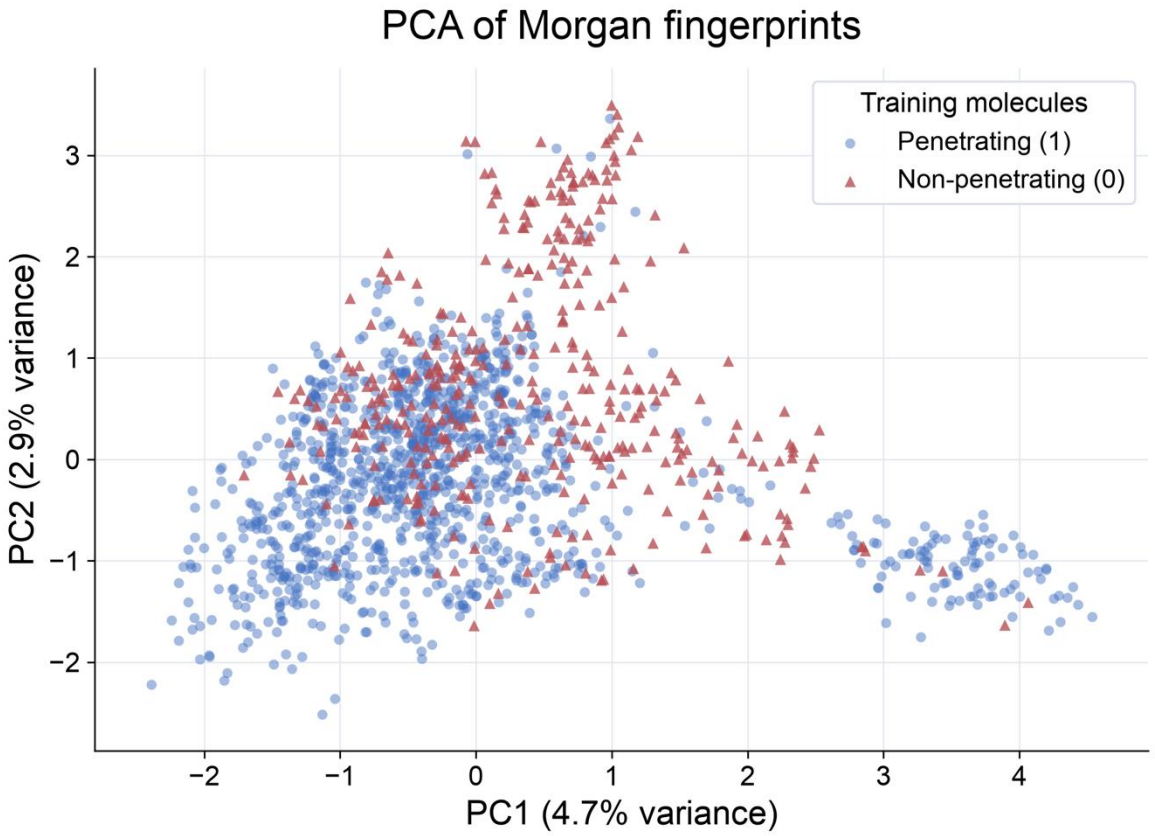
Building the Feature Matrix and Target Vector

Feature Matrix, X: **2048-bit** morgan fingerprint, radius of **2**. **1915** rows.

PCA to keep 90% of variances

PCA fitting only on **training data!**





Interactive Python Time

27_molecular_property_prediction.ipynb

In this lecture, we learned

- *Construct molecular inputs and experimental targets*
- *Train a regression model and a classification model*
- *Select model settings using training cross-validation*

Next lecture:

- *Molecules as graphs*
- *Graph representations: nodes and edges*
- *NetworkX*